

Drying Out a Crawlspace

by Terry Brennan

Vented crawlspaces are standard construction practice, but they have been linked to ice dams, persistent indoor condensation in the winter, and some fairly severe health problems.

In the fall of 1998, a homeowner called my company, Camroden Associates, because she was having respiratory problems. Her symptoms included clogged sinuses, sore throats, and congestion—typical hay fever symptoms.

When folks call, I usually try to let them tell their story with as little help from me as possible. That old adage of “Take the cotton out of your ears and put it in your mouth” works pretty well, when I remember to do it. I put the information I get into mental bins labeled “occupant information,” “beneficial or aggravating transport factors,” and “contaminant information.”

Occupant Information

The occupant information in this case was as follows: The occupant is severely allergic to mold and is being treated by an allergist; she smells moldy odors in the house; and her home is built over a vented crawlspace that smells particularly musty. She and her husband are just finishing a 400 ft² addition, which is also over a crawlspace. However, the new crawlspace is insulated, has exterior drainage, and has a ground cover.

I visited the house in October 1998. It's located in central New York, which has a cold, wet climate (7,000 degree-

days, 50 inches of rainfall, and 300 overcast days per year). While visiting the house, I confirmed the occupant's observations.

Transport Factors

An inspection of the house revealed the following information about the transport factors:

- The house is a single-story, wood frame structure with a concrete block foundation.
- It has a vented crawlspace with a floor that is about two-thirds exposed earth and about one-third exposed bedrock. About one-tenth of the

exposed-earth area was damp to the touch (the clay formed balls and ribbons in my hand). There were erosion channels cut in the earth by liquid water entering the high side of the crawlspace and running to the low side. The crawlspace air was 65°F with 80% relative humidity (RH), while the air in the living space and outdoors was 72°F with 45% RH.

- The house has a plywood subfloor over 2 x 10 floor joists.
- No combustion appliances are located in the crawlspace. The house is heated with a warm-air furnace, which is upstairs. However, both return and supply ductwork run through the crawlspace. Operating the air handler with the crawlspace vent screens closed depressurized the crawlspace by 1 Pascal (Pa).
- There is no floor insulation.
- The house has vertical board-and-batten painted cedar siding.
- The double-pane windows are air-vented, with vents between the double glass and the outside air.

- The roofing consists of cedar shake shingles on tongue-and-groove sheathing.
- There is cellulose insulation in the attic, and there are mineral wool batts in the walls.
- The crawlspace beneath the new addition is insulated with extruded styrene on the exterior walls, is air sealed, has a polyethylene groundcover, and has an exterior drainage system.

Discoveries

In a case like this, I need to identify the extent of the moisture problem. Through diagnostic evaluations, I made the following discoveries:

- A moisture content (MC) survey with an electronic moisture meter indi-



The existing clay drain tiles were found to be completely filled with soil.

cated that the rim joists were saturated with a MC greater than 20%, while the framing in the rest of the crawlspace was around 15%–20% MC. (A normal MC level for wood products is generally 11%–12%.)

- The concrete block foundation was not damp to the touch, but was close to saturation on the moisture content meter.
- The MC of the framing in the living space and in the attic was all 10% or less.
- I blocked off the crawlspace vents and then used a small fan to depressurize the space, resulting in a flow rate of 230 CFM at 2 Pa. Clearly, even with the blocked vents, the crawlspace was relatively leaky and would need air sealing.
- In a previous effort to dry out the crawlspace, the HVAC contractor had installed an oversized DX air conditioning unit in the new addition, which also blew conditioned air (air conditioned in the summer and heated in the winter) into the crawlspace.

Contaminant Information

To identify the extent of fungal contamination, I inspected the area with the naked eye, a hand lens, and a microscope. A glance at the psychrometric chart showed that the crawlspace air had more water vapor than the air in the living space and out-

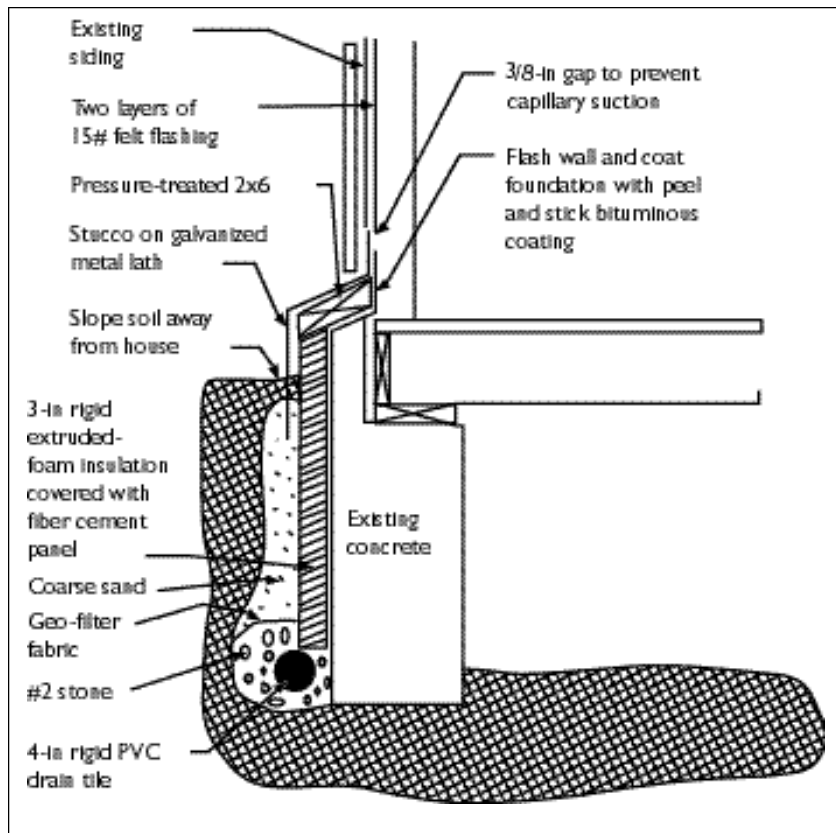


Figure 1. A new drainage system, combined with dampproofing and insulation, will keep moisture out of the crawlspace.

doors, which was evidence that there was a source of water in the crawlspace. The crawlspace also had an obvious moldy odor.

Many of the materials in the crawlspace were damp enough to support fungal growth, but there were very few colonies of fungus large enough to be seen with the naked eye. However, when I used magnification, I could see that there were numerous colonies on the rim joists, on debris on the crawlspace floor, and in the soil itself. This is a common condition—just enough moisture to support a large number of tiny colonies. I generally don't take samples to try to identify the type of mold, because I believe that we need to get rid of mold no matter what type it is. I also think that all workers in a moldy crawlspace must wear protection.

Recommendations

Based on this information, I made the following five recommendations to the client:

1. Install or rehabilitate the existing exterior perimeter drainage.
2. Insulate and air seal the existing crawlspace wall.
3. Air seal the return ductwork in the crawlspace and the floor deck between the living space and the crawlspace.
4. Close off the supply ducts that open into the crawlspace.
5. Install an exhaust fan to depressurize the crawlspace.

The exterior drainage will prevent rainwater from entering the crawlspace. Air sealing the crawlspace wall will prevent water vapor from entering the crawlspace as part of infiltrating outdoor air during the warm, humid months. Insulating the crawlspace walls will raise the crawlspace temperature during the heating season. This in turn will lower the relative humidity in the crawlspace, increase thermal comfort in the house, and increase the thermal efficiency of the supply duct system.

Sealing the return ductwork in the crawlspace will prevent the crawlspace air from moving through the ductwork into the living space. Sealing the floor deck will help to keep crawlspace air out of the house and will allow a smaller fan to be used to depressurize the crawlspace. The supply duct open-

ings must be closed off, or the crawlspace will be pressurized, which will blow crawlspace air into the occupied space. Also, the fact that supply air is entering the crawlspace means there is more return air leaving the house than supply air entering it; therefore, the house is being depressurized, making the problem worse. For this reason, supply duct leaks need to be sealed also.

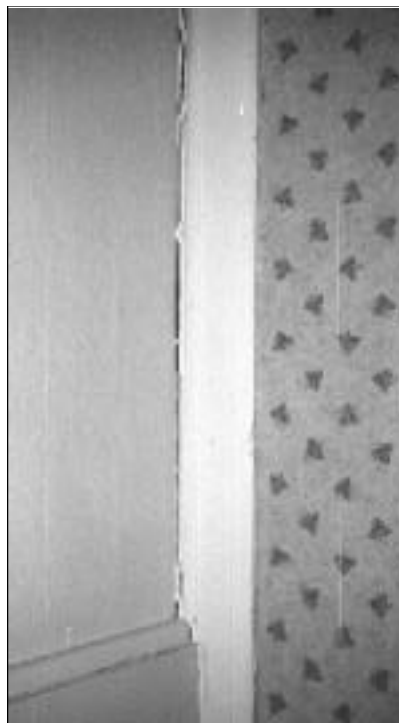
Using an exhaust fan to depressurize the crawlspace will reverse the home's habitual air flow. Without the fan, air moves from the crawlspace to the conditioned space in winter, because of the stack effect. With the fan, air will be drawn down from the occupied space to the space that houses a possible source of contaminants. With a 90W fan running, the homeowners will have additional annual energy costs, but they will be saving a fair amount of energy by insulating their crawlspace and sealing their ductwork. They will also be getting consistent ventilation at a rate of 80 CFM.

With the homeowners' approval, the following measures were taken:

- I installed an exhaust fan in the crawlspace during the first visit, and within a week made it a permanent installation.
- I sealed the crawlspace vents. The crawlspace was running about 2 Pa negative. A thorough cleaning of all the soft materials in the occupied space—including books and carpets—was undertaken to remove accumulated allergens from the household. This was a critical step in alleviating the occupant's symptoms.
- The exterior of the foundation was excavated by hand to conserve flowers and shrubs. The existing clay tiles, which were used to channel water around the house in the original drainage system, were found to be completely filled with soil. Damp-proofing, insulation, and new drainage were installed (see Figure 1). The ductwork and penetrations through the floor were sealed.

Results

The occupant's symptoms have remained at a reduced level since the work was completed, but they have not



The wall trim, shrinking due to less humidity in the home, is leaving a visible gap.

completely disappeared. They currently seem to be linked to heavy rainfall. Because around 150 CFM of air is still leaking into the ducts when the air handler runs, a greater effort to seal off the return air leaks could be made.

In addition to reducing the occupant's symptoms, depressurizing the crawlspace appears to have eliminated perennial ice dam problems and wintertime condensation on the windows. The ice dam and condensation problems had been ongoing since the owners moved in five years ago. The homeowner is now noticing cracks at seams in the gypsum board and shrinking of wooden items such as doors and trims. The cracks in the gypsum board appeared to be the first since the home was built, in the early 1950s. Apparently the crawlspace had been humidifying the house all those years. The homeowners may now need to add some moisture to the home during the coldest months, because they currently have wintertime humidities of around 20%. ■

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